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Object-oriented programming organizes software around objects, classes, inheritance, encapsulation, polymorphism, and abstraction. Object-oriented programming organizes software around objects, classes, inheritance, encapsulation, polymorphism, and abstraction. Object-oriented programming organizes software around objects, classes, inheritance, encapsulation, polymorphism, and abstraction. Object-oriented programming organizes software around objects, classes, inheritance, encapsulation, polymorphism, and abstraction. Object-oriented programming organizes software around objects, classes, inheritance, encapsulation, polymorphism, and abstraction. Object-oriented programming organizes software around objects, classes, inheritance, encapsulation, polymorphism, and abstraction. Object-oriented programming organizes software around objects, classes, inheritance, encapsulation, polymorphism, and abstraction. Object-oriented programming organizes software around objects, classes, inheritance, encapsulation, polymorphism, and abstraction. Example 1: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.

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Chapter 4: Algorithms

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Node.js enables server-side JavaScript with an event-driven, non-blocking architecture. Node.js enables server-side JavaScript with an event-driven, non-blocking architecture. Node.js enables server-side JavaScript with an event-driven, non-blocking architecture. Node.js enables server-side JavaScript with an event-driven, non-blocking architecture. Node.js enables server-side JavaScript with an event-driven, non-blocking architecture. Node.js enables server-side JavaScript with an event-driven, non-blocking architecture. Node.js enables server-side JavaScript with an event-driven, non-blocking architecture. Node.js enables server-side JavaScript with an event-driven, non-blocking architecture. Node.js enables server-side JavaScript with an event-driven, non-blocking architecture. Example 8: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.

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Chapter 11: Kubernetes

Kubernetes orchestrates containers, providing scaling, self-healing, and rolling deployments.
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Chapter 12: Cloud Computing

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Example 3: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.

Cloud platforms provide on-demand infrastructure, networking, storage, and managed services.
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Chapter 13: Machine Learning

Machine learning enables systems to learn patterns from data for prediction and classification.
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Chapter 15: Embeddings

Embeddings convert text into dense vectors that preserve semantic meaning. Embeddings convert text into dense vectors that preserve semantic meaning. Embeddings convert text into dense vectors that preserve semantic meaning. Embeddings convert text into dense vectors that preserve semantic meaning. Embeddings convert text into dense vectors that preserve semantic meaning. Embeddings convert text into dense vectors that preserve semantic meaning. Embeddings convert text into dense vectors that preserve semantic meaning. Example 1: This section discusses concepts, best

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Chapter 16: Vector Databases

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RAG retrieves relevant knowledge before sending context to an LLM, improving factual accuracy.
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Chapter 18: System Design

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Chapter 21: Python Programming

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Object-oriented programming organizes software around objects, classes, inheritance, encapsulation, polymorphism, and abstraction. Object-oriented programming organizes software around objects, classes, inheritance, encapsulation, polymorphism, and abstraction. Object-oriented programming organizes software around objects, classes, inheritance, encapsulation, polymorphism, and abstraction. Object-oriented programming organizes software around objects, classes, inheritance, encapsulation, polymorphism, and abstraction. Object-oriented programming organizes software around objects, classes, inheritance, encapsulation, polymorphism, and abstraction. Object-oriented programming organizes software around objects, classes, inheritance, encapsulation, polymorphism, and abstraction. Object-oriented programming organizes software around objects, classes, inheritance, encapsulation, polymorphism, and abstraction. Object-oriented programming organizes software around objects, classes, inheritance, encapsulation, polymorphism, and abstraction. Example 1: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.

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Arrays, linked lists, stacks, queues, trees, graphs, and hash tables are fundamental data structures.
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 Example 1: This section discusses concepts, best practices, implementation details, trade-offs,
 real-world scenarios, and interview questions.

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Chapter 24: Algorithms

Sorting, searching, recursion, dynamic programming, and graph algorithms solve computational problems efficiently. Sorting, searching, recursion, dynamic programming, and graph algorithms solve computational problems efficiently. Sorting, searching, recursion, dynamic programming, and graph algorithms solve computational problems efficiently. Sorting, searching, recursion, dynamic programming, and graph algorithms solve computational problems efficiently. Sorting, searching, recursion, dynamic programming, and graph algorithms solve computational problems efficiently. Sorting, searching, recursion, dynamic programming, and graph algorithms solve computational problems efficiently. Sorting, searching, recursion, dynamic programming, and graph algorithms solve computational problems efficiently. Sorting, searching, recursion, dynamic programming, and graph algorithms solve computational problems efficiently. Example 1: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.

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[illegible]

JavaScript powers interactive web applications and runs in browsers and on servers through Node.js. JavaScript powers interactive web applications and runs in browsers and on servers through Node.js. JavaScript powers interactive web applications and runs in browsers and on servers through Node.js. JavaScript powers interactive web applications and runs in browsers and on servers through Node.js. JavaScript powers interactive web applications and runs in browsers and on servers through Node.js. JavaScript powers interactive web applications and runs in browsers and on servers through Node.js. JavaScript powers interactive web applications and runs in browsers and on servers through Node.js. JavaScript powers interactive web applications and runs in browsers and on servers through Node.js. Example 7: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.

Chapter 26: React

React builds component-based user interfaces using state, props, hooks, and virtual DOM. React builds component-based user interfaces using state, props, hooks, and virtual DOM. React builds component-based user interfaces using state, props, hooks, and virtual DOM. React builds component-based user interfaces using state, props, hooks, and virtual DOM. React builds component-based user interfaces using state, props, hooks, and virtual DOM. React builds component-based user interfaces using state, props, hooks, and virtual DOM. React builds component-based user interfaces using state, props, hooks, and virtual DOM. React builds component-based user interfaces using state, props, hooks, and virtual DOM. Example 2: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.

[illegible]

Relational databases use SQL while NoSQL databases trade strict schemas for flexibility.
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 7: This section discusses concepts, best practices, implementation details, trade-offs, real-world
 scenarios, and interview questions.

Chapter 30: Docker

Docker packages applications into portable containers with consistent runtime environments.
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Example 6: This section discusses concepts, best practices, implementation details, trade-offs,
real-world scenarios, and interview questions.

[illegible]

[illegible][illegible][illegible]

Chapter 32: Cloud Computing

[illegible][illegible]

Cloud platforms provide on-demand infrastructure, networking, storage, and managed services.
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Chapter 33: Machine Learning

Machine learning enables systems to learn patterns from data for prediction and classification.
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Example 4: This section discusses concepts, best practices, implementation details, trade-offs,
real-world scenarios, and interview questions.

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Chapter 34: Large Language Models

LLMs are transformer-based neural networks trained on massive text corpora. LLMs are transformer-based neural networks trained on massive text corpora. LLMs are transformer-based neural networks trained on massive text corpora. LLMs are transformer-based neural networks trained on massive text corpora. LLMs are transformer-based neural networks trained on massive text corpora. LLMs are transformer-based neural networks trained on massive text corpora. LLMs are transformer-based neural networks trained on massive text corpora. LLMs are transformer-based neural networks trained on massive text corpora. LLMs are transformer-based neural networks trained on massive text corpora. Example 2: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.

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Embeddings convert text into dense vectors that preserve semantic meaning. Embeddings convert text into dense vectors that preserve semantic meaning. Embeddings convert text into dense vectors that preserve semantic meaning. Embeddings convert text into dense vectors that preserve semantic meaning. Embeddings convert text into dense vectors that preserve semantic meaning. Embeddings convert text into dense vectors that preserve semantic meaning. Embeddings convert text into dense vectors that preserve semantic meaning. Embeddings convert text into dense vectors that preserve semantic meaning. Example 8: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.

Vector databases store embeddings and perform similarity search. Vector databases store embeddings and perform similarity search. Vector databases store embeddings and perform similarity search. Vector databases store embeddings and perform similarity search. Vector databases store embeddings and perform similarity search. Vector databases store embeddings and perform similarity search. Vector databases store embeddings and perform similarity search. Vector databases store embeddings and perform similarity search. Example 1: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.

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RAG retrieves relevant knowledge before sending context to an LLM, improving factual accuracy. Example 3: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.

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Example 5: This section discusses concepts, best practices, implementation details, trade-offs,
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Example 7: This section discusses concepts, best practices, implementation details, trade-offs,
real-world scenarios, and interview questions.

Scalable systems emphasize reliability, availability, caching, load balancing, and observability.
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Scalable systems emphasize reliability, availability, caching, load balancing, and observability.
Example 7: This section discusses concepts, best practices, implementation details, trade-offs,
real-world scenarios, and interview questions.

Chapter 39: REST APIs

REST uses HTTP methods such as GET, POST, PUT, and DELETE to expose resources. REST uses HTTP methods such as GET, POST, PUT, and DELETE to expose resources. REST uses HTTP methods such as GET, POST, PUT, and DELETE to expose resources. REST uses HTTP methods such as GET, POST, PUT, and DELETE to expose resources. REST uses HTTP methods such as GET, POST, PUT, and DELETE to expose resources. REST uses HTTP methods such as GET, POST, PUT, and DELETE to expose resources. REST uses HTTP methods such as GET, POST, PUT, and DELETE to expose resources. REST uses HTTP methods such as GET, POST, PUT, and DELETE to expose resources. Example 2: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.

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DevOps combines automation, CI/CD, monitoring, infrastructure as code, and collaboration.
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Example 5: This section discusses concepts, best practices, implementation details, trade-offs,
real-world scenarios, and interview questions.

[illegible][illegible]

Chapter 41: Python Programming

Python is a versatile programming language used for web development, automation, AI, and data science. It emphasizes readability and has a rich ecosystem of libraries.

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Example 2: This section discusses concepts, best practices, implementation details,

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Chapter 42: Object-Oriented Programming

[illegible]

Arrays, linked lists, stacks, queues, trees, graphs, and hash tables are fundamental data structures.
 Arrays, linked lists, stacks, queues, trees, graphs, and hash tables are fundamental data structures.
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 Example 8: This section discusses concepts, best practices, implementation details, trade-offs,
 real-world scenarios, and interview questions.

Sorting, searching, recursion, dynamic programming, and graph algorithms solve computational problems efficiently. Sorting, searching, recursion, dynamic programming, and graph algorithms solve computational problems efficiently. Sorting, searching, recursion, dynamic programming, and graph algorithms solve computational problems efficiently. Sorting, searching, recursion, dynamic programming, and graph algorithms solve computational problems efficiently. Sorting, searching, recursion, dynamic programming, and graph algorithms solve computational problems efficiently. Sorting, searching, recursion, dynamic programming, and graph algorithms solve computational problems efficiently. Sorting, searching, recursion, dynamic programming, and graph algorithms solve computational problems efficiently. Sorting, searching, recursion, dynamic programming, and graph algorithms solve computational problems efficiently. Example 1: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.

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JavaScript powers interactive web applications and runs in browsers and on servers through Node.js. JavaScript powers interactive web applications and runs in browsers and on servers through Node.js. JavaScript powers interactive web applications and runs in browsers and on servers through Node.js. JavaScript powers interactive web applications and runs in browsers and on servers through Node.js. JavaScript powers interactive web applications and runs in browsers and on servers through Node.js. JavaScript powers interactive web applications and runs in browsers and on servers through Node.js. JavaScript powers interactive web applications and runs in browsers and on servers through Node.js. JavaScript powers interactive web applications and runs in browsers and on servers through Node.js. Example 6: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.

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React builds component-based user interfaces using state, props, hooks, and virtual DOM. React builds component-based user interfaces using state, props, hooks, and virtual DOM. React builds component-based user interfaces using state, props, hooks, and virtual DOM. React builds component-based user interfaces using state, props, hooks, and virtual DOM. React builds component-based user interfaces using state, props, hooks, and virtual DOM. React builds component-based user interfaces using state, props, hooks, and virtual DOM. React builds component-based user interfaces using state, props, hooks, and virtual DOM. React builds component-based user interfaces using state, props, hooks, and virtual DOM. Example 1: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.

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Docker packages applications into portable containers with consistent runtime environments.
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Example 3: This section discusses concepts, best practices, implementation details, trade-offs,
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Chapter 52: Cloud Computing

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Example 1: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.

Example 2: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.

Example 3: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.

Example 4: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.

Example 5: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.

Example 6: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.

Machine learning enables systems to learn patterns from data for prediction and classification.
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Example 7: This section discusses concepts, best practices, implementation details, trade-offs,
real-world scenarios, and interview questions.

Chapter 54: Large Language Models

LLMs are transformer-based neural networks trained on massive text corpora. LLMs are
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Embeddings convert text into dense vectors that preserve semantic meaning. Embeddings convert text into dense vectors that preserve semantic meaning. Embeddings convert text into dense vectors that preserve semantic meaning. Embeddings convert text into dense vectors that preserve semantic meaning. Embeddings convert text into dense vectors that preserve semantic meaning. Embeddings convert text into dense vectors that preserve semantic meaning. Embeddings convert text into dense vectors that preserve semantic meaning. Example 1: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.

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Chapter 56: Vector Databases

Vector databases store embeddings and perform similarity search. Vector databases store embeddings and perform similarity search. Vector databases store embeddings and perform similarity search. Vector databases store embeddings and perform similarity search. Vector databases store embeddings and perform similarity search. Vector databases store embeddings and perform similarity search. Vector databases store embeddings and perform similarity search. Example 2: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.

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[illegible][illegible]

RAG retrieves relevant knowledge before sending context to an LLM, improving factual accuracy.
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Example 2: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.

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 Example 6: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.

Chapter 58: System Design

Scalable systems emphasize reliability, availability, caching, load balancing, and observability.
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 Example 8: This section discusses concepts, best practices, implementation details, trade-offs,
 real-world scenarios, and interview questions.

REST uses HTTP methods such as GET, POST, PUT, and DELETE to expose resources. REST uses HTTP methods such as GET, POST, PUT, and DELETE to expose resources. REST uses HTTP methods such as GET, POST, PUT, and DELETE to expose resources. REST uses HTTP methods such as GET, POST, PUT, and DELETE to expose resources. REST uses HTTP methods such as GET, POST, PUT, and DELETE to expose resources. REST uses HTTP methods such as GET, POST, PUT, and DELETE to expose resources. REST uses HTTP methods such as GET, POST, PUT, and DELETE to expose resources. REST uses HTTP methods such as GET, POST, PUT, and DELETE to expose resources. REST uses HTTP methods such as GET, POST, PUT, and DELETE to expose resources. Example 1: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.

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Chapter 60: DevOps

Example 1: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.

Example 2: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.

Example 3: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.

Example 4: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.

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Example 5: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.

DevOps combines automation, CI/CD, monitoring, infrastructure as code, and collaboration.
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Example 8: This section discusses concepts, best practices, implementation details, trade-offs, real-world scenarios, and interview questions.